

SmolVLA Q10 critic and corrector

Pipeline, architecture, learning objective, and current evidence

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Scope. The current SmolVLA Q10 model is a *critic*: it scores an already proposed ten-action chunk. Candidate reranking is the nearest-term *corrector* experiment. The repository also contains an older, distinct PCP gradient corrector, which nudges a denoising estimate inside P&P. These mechanisms must not be treated as a single validated system.

1. The quantity the critic should learn

At a LIBERO decision boundary, let x_t denote the observation, task instruction, robot state, and remaining episode budget. Let $A_t = (a_t, \dots, a_{t+9}) \in \mathbb{R}^{10 \times 7}$ be the *ten actions actually executed*. The target is the success probability after one intervention and a fixed continuation policy π :

$$Q_{10}^\pi(x_t, A_t) = \Pr(Y = 1 \mid x_t, \text{do}(A_t), \text{then frozen SmolVLA P\&P } \pi). \quad (1)$$

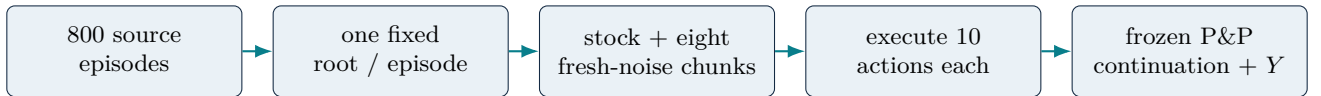
Here Y is binary terminal task success. The policy and its rollout procedure define the estimand: changing the continuation policy changes Q . This is *policy evaluation* over candidate chunks, not an optimal-action Q^* or a reward model for arbitrary actions.

For a recorded ten-step transition, the one-step Bellman target is

$$y_t = r_t + (1 - d_t) \bar{Q}(x_{t+10}, A_{t+10}^\pi), \quad \gamma = 1, \quad (2)$$

where r_t records success in that interval, d_t is termination, and A_{t+10}^π is the *recorded* next chunk under frozen π . The target network $\bar{Q}$ is an exponential moving average. There is no $\max_A$: inserting one would train a different control objective. Remaining time belongs in x_t because the task has a finite step limit.

2. Data-generation pipeline

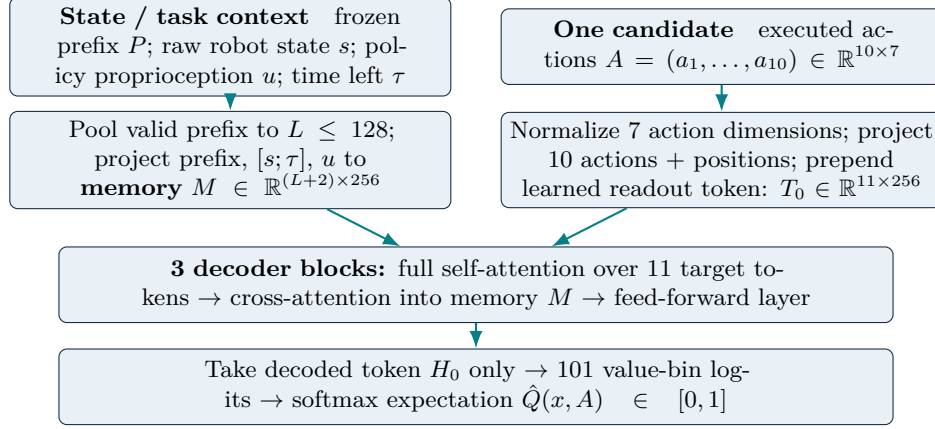


The source cohort contains 800 distinct episodes at indices 10–29. Root locations were fixed in an immutable manifest before the new candidate outcomes: 65% were selected by a high U_{10} uncertainty score, 35% uniformly. The v4 fresh8 tree defines nine candidates at the *same state*: the exact stored source chunk A_0 , plus $A_1, \dots, A_8$ generated from fresh initial noise using P&P. Each chunk is executed for ten actions; the same frozen P&P continuation and future seed then produce an observed success/failure. Complete v3 roots can reuse their four existing fresh-noise branch artifacts; the remaining four are collected in v4. The four v3 perturbation-stream branches are excluded from v4.

The worker writes candidate metadata, chunk artifacts, observations, and continuations to Supabase. A separate snapshot/cache job validates complete trees and builds root records and ten-step transition windows. Roots and all nine branches stay together in an 80/20 train/validation split, stratified by suite and source outcome. **Unit of independent generalization: root state, not branch.** Nine correlated outcomes at one root are particularly valuable for same-state ranking, but 800 trees are 800 states, not 7,200 independent states.

3. Architecture: one state and one candidate chunk in, one score out

The critic is a *separate* action-conditioned network. It receives cached SmolVLA context and an already generated 10×7 chunk; it does not run the VLA action generator during critic training. Two token streams meet in a Transformer decoder:



Context stream. $P \in \mathbb{R}^{L_0 \times d_p}$ is the stored visual/language prefix embedding at this decision boundary, with a validity mask. Valid tokens are kept in order; if $L_0 > 128$, adaptive average pooling compresses them to 128. Each pooled token passes through a learned layer norm and linear projection to width 256. The raw robot vector augmented with remaining episode fraction, $[s; \tau]$, and the SmolVLA policy proprioception vector u each receive their *own* learned norm/linear projection and are appended as two memory tokens. Thus, schematically,

$$M = [\phi_P(\tilde{P}_1), \dots, \phi_P(\tilde{P}_L), \phi_s([s; \tau]), \phi_u(u)] \in \mathbb{R}^{(L+2) \times 256}. \quad (3)$$

The upstream SmolVLA features are frozen/cached; these three critic-side projections *are trained*. The raw robot state and policy proprioception are separate inputs even if some information overlaps.

Action stream and readout. Action means $\mu \in \mathbb{R}^7$ and standard deviations $\sigma \in \mathbb{R}^7$ are fitted on training-root candidates only. For action position i , $e_i = \text{LN}(W_a((a_i - \mu)/\sigma) + b_a) + p_i$, where p_i is a learned position vector. A learned vector c_{RL} is prepended: $T_0 = [c_{\text{RL}}, e_1, \dots, e_{10}]$. The “RL token” is simply a learned *readout slot*, analogous to a classification token; it is not an extra reward observation or a separate RL policy. There is no causal target mask: in each of the three decoder layers it can attend to all ten candidate actions, while cross-attention reads the state/task memory. The final prediction uses only that token:

$$H = D_\theta^{(3)}(T_0; M), \quad z = W_v H_0 + b_v, \quad \hat{Q}(x, A) = \sum_{k=0}^{100} \frac{k}{100} \text{softmax}(z)_k. \quad (4)$$

The decoder has width 256, 8 heads, FFN width 1024, and dropout 0.20; with a 512-wide prefix, the critic has about 3.33 million trainable parameters. For nine candidates at one root, the code repeats the *same* context nine times and scores each chunk independently: candidates do not attend to one another. The 101-bin output is a modeling choice, not evidence of 101 observed return values; root outcomes are binary.

4. Two training arms on one immutable snapshot

Root Monte Carlo (MC). For each root j and candidate i , the full continuation gives $Y_{ji} \in \{0, 1\}$. The implemented loss applies binary cross-entropy to the *expected value* of the 101-bin head:

$$\mathcal{L}_{\text{MC}} = -\frac{1}{9B} \sum_{j=1}^B \sum_{i=0}^8 [Y_{ji} \log \hat{q}_{ji} + (1 - Y_{ji}) \log(1 - \hat{q}_{ji})]. \quad (5)$$

Training samples roots and evaluates all nine same-state candidates. This directly supervises the proposed intervention but each label reflects one continuation random stream.

Tree temporal difference (TD). The cache provides ten-action windows through the branch continuation. The implemented target is $y_t = \text{clip}_{[0,1]} \{r_t + (1 - d_t)\bar{q}_{t+10}\}$. The scalar y_t is spread over nearby value bins using a narrow Gaussian target $h_k(y_t)$, followed by categorical cross-entropy:

$$\mathcal{L}_{\text{TD}} = -\frac{1}{B} \sum_t \sum_{k=0}^{100} h_k(y_t) \log p_{\theta,k}(x_t, A_t), \quad \bar{\theta} \leftarrow (1 - 0.005)\bar{\theta} + 0.005\theta. \quad (6)$$

TD uses more boundaries, but bootstraps and may spend most updates on downstream continuation actions rather than fork choices. A single sigmoid head with BCE against Y for MC or soft target $y_t \in [0, 1]$ for TD is a clean, *not-yet-run* ablation. Both forms seek a success probability; the bins are not required by the Bellman equation.

Both arms currently use AdamW (10^{-4} initial learning rate, cosine schedule, 100 warmup updates), batch size 8 roots for MC or 8 sampled TD branches with 8 windows each, 2,000 updates, gradient clipping at 1, and periodic held-out evaluation. Q training runs on cached features and does not require running SmolVLA or LIBERO on the training GPU.

5. What “corrector” means here

Proposed, conservative candidate selector. At deployment, generate A_0 (stock) and fresh P&P candidates $A_1, \dots, A_8$ at one observation, then score them with $\hat{Q}$. A stock-preserving rule is

$$i^* = \arg \max_{i \geq 1} \hat{Q}(x, A_i), \quad A_{\text{chosen}} = \begin{cases} A_{i^*}, & \hat{Q}(x, A_{i^*}) - \hat{Q}(x, A_0) > \tau, \\ A_0, & \text{otherwise.} \end{cases} \quad (7)$$

The threshold τ must be tuned on a separate development set and evaluated on fresh roots. This rule *selects* an already generated on-distribution chunk. It does not modify the chunk. A future action optimizer could maximize $\hat{Q}(x, A) - \lambda \|A - A_0\|^2$, possibly with P&P projection, but it risks exploiting errors off the training distribution and needs new simulator evaluations.

Historical PCP gradient corrector. The separate `pnp/pcp.py` path has a three-layer residual MLP over a flattened denoising estimate, observation embedding, and chunk-position/noise-level features. When its calibrated success score falls below a gate, it applies $z' = z + \lambda \nabla_z \hat{Q}_{\text{PCP}}$, then re-noises the result within P&P. This is an actual action-changing corrector, but it is *not* the current SmolVLA Q10 transformer, its training data, or a validated consequence of the fresh8 experiment.

6. Results, limits, and immediate decision

On the completed **v3 480-tree snapshot** (384 training roots, 96 held-out roots; original + four fresh + four perturbation candidates), stock succeeded on 66/96 validation roots; an oracle over recorded candidates would succeed on 74/96. After 2,000 updates, ungated root-MC argmax selected 60/96 (0 rescues, 6 spoils; within-root pair accuracy 0.562). TD argmax selected 56/96 (2 rescues, 12 spoils; pair accuracy 0.475). Thus the observed critic-based argmax *degraded* success in this holdout. These results do not yet establish a useful deployed corrector.

The intended **v4 fresh8** experiment has 800 possible roots and nine fresh-only candidates per root; the code and three Colab workers exist. The three one-tree smoke runs shown in the conversation completed successfully, but they are not evidence that all 800 trees are complete. Freeze a complete v4 snapshot before training and report its actual root count. Then use nested training-root subsets with a fixed validation set to measure scaling. Evaluate stock, oracle, selected success, rescues, spoils, mixed-root pair accuracy, calibration, and per-suite effects. Reserve unseen episode indices for a final test; the existing 10–29 cohort and single future seed per branch limit external validity.

7. Interpretation and next controlled comparisons

The central test is whether the model can *rank actions at the same root*, not merely predict whether a task or scene is easy. An observation-only shortcut can achieve useful calibration while failing to order the nine candidates. Report a within-root pair metric only on mixed roots, and compare selection to the stock action at each identical root.

The controlled comparisons are: (i) nested 100/200/400/~640 training roots with one fixed holdout, (ii) the 101-bin decoder against a scalar sigmoid and a smaller action-conditioned head, (iii) root MC against TD with root-balanced window sampling, and (iv) stock-only, ungated argmax, and a threshold tuned away from the final test. Change one factor at a time. For a genuine action-changing corrector, score improvement on *newly generated and executed* chunks. Offline selection within the recorded nine candidates cannot establish that gradient ascent in action space remains on the policy’s support. Conversely, the fresh8 candidates directly match the P&P proposal distribution the selector would encounter, which is why they are the clean first test.

Repository basis. `pnnp/smolvla_fresh8_collection.py`; `pnnp/smolvla_tree_bellman_finetune.py`;
`pnnp/qplanning_critic/model.py`; `pnnp/smolvla_success_critic.py`; `pnnp/pcp.py`;
`SMOLVLA_Q10_DIAGNOSIS_2026-09-26.md`. Status and results reflect this repository snapshot and its documented Modal run; this report did not perform a new training run.